

ICSA 2026 SHORT COURSE

Lecture 3: Covariate shift and density-ratio reweighting

Statistical Foundations of Transfer Learning

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Overview

§3.1: Covariate shift

§3.2: Is covariate shift adaptable for free?

§3.3: The reweighting method

Numerical illustration: importance weighting

§3.4: Density-ratio estimation

References

§3.1: Covariate shift

Covariate shift, formally

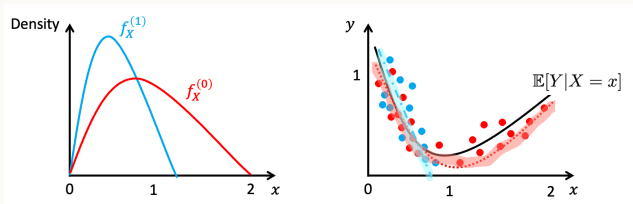
In $L2$ the covariate-shift gap was a fixed penalty $2d_{TV}$ we could only bound; this lecture shows importance weighting removes it and recovers oracle rates.

- Source $(X, Y) \sim \mathbb{P}^{(1)}$, target $(X, Y) \sim \mathbb{P}^{(0)}$, on $\mathcal{X} \times \mathcal{Y}$.

- **Covariate shift:**

$$\mathbb{P}_X^{(0)} \neq \mathbb{P}_X^{(1)}, \quad \mathbb{P}_{Y|X}^{(0)} = \mathbb{P}_{Y|X}^{(1)} \quad (\text{a.s. } \mathbb{P}_X^{(0)}).$$

- Equivalent: there exists a regression / decision function f^* shared by source and target; only the feature distribution changes.



Three motivating scenarios

- **Self-driving cars:** training routes $\subsetneq$ deployment routes. Same physics; new lighting, weather, traffic distributions.
- **Medical imaging:** Alzheimer's from MRI: a research-study scanner (source) vs. a community-hospital scanner (target) — same disease, shifted images (our L1 running example).
- **A/B testing for ML:** the production traffic mix shifts after a model is shipped, but the conditional model of $Y \mid X$ is unchanged.

Why special: in all three, the *physics/biology/intent* is fixed — only the *distribution of situations* changes. That is covariate shift.

Density ratio: the central object

Define the source-to-target density ratio

$$w(\mathbf{x}) = \frac{d\mathbb{P}_X^{(0)}}{d\mathbb{P}_X^{(1)}}(\mathbf{x}),$$

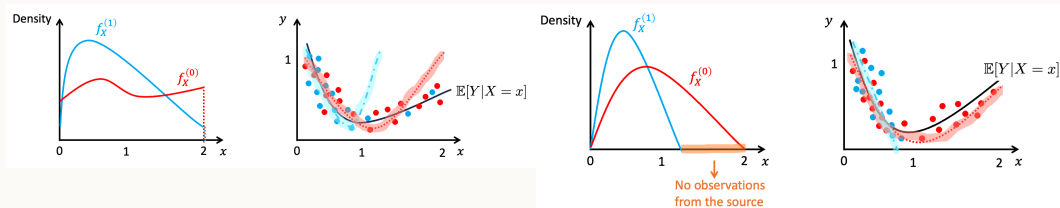
assuming $\mathbb{P}_X^{(0)} \ll \mathbb{P}_X^{(1)}$ (target absolutely continuous w.r.t. source).

- Intuition: $w(\mathbf{x}) > 1$ at points the target “cares about” more than the source.
- Failure case: $\sup_{\mathbf{x}} w(\mathbf{x}) = \infty$. Then any source-only estimator can be fooled.
- Key fact: for any integrable g , $\mathbb{E}_{\mathbb{P}^{(0)}}[g(X)] = \mathbb{E}_{\mathbb{P}^{(1)}}[w(X)g(X)]$.
- *Think of it as:* reweighting source points by w lets you “pretend” you sampled from the target — importance sampling over $\mathcal{X}$.

Bounded vs. unbounded density ratios

Bounded w (well-behaved)

Unbounded w (failure)



Bounded w + small effective sample size $n_1/\|w\|_\infty \Rightarrow$ usable transfer.

Unbounded w : a single source point in a region the source rarely visits but the target loves gets an astronomically large weight — one observation dominates the empirical risk.

Check-in: what defines covariate shift?

In which of the following scenarios is the assumption $\mathbb{P}_{Y|X}^{(0)} = \mathbb{P}_{Y|X}^{(1)}$ most plausible?

- (a) Survey of US households 2010 vs. 2020: X is income, Y is voting preference.
- (b) Camera sees same kind of objects under new lighting; X is the pixel array, Y is the object class.
- (c) Hospital A's MRI vs. Hospital B's CT scan; Y is whether the patient has tumor.
- (d) Pre- vs. post-pandemic search queries; Y is whether to click a particular ad.

Answer

(b). Same physical objects under different lighting – the underlying object-given-pixel mapping is unchanged. (a) and (d) involve a shifting world (people change their preferences), and (c) doesn't even share the feature space (X is different).

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§3.2: Is covariate shift adaptable for free?

When ERM survives covariate shift unaided

Consider parametric MLE under a **well-specified** model: $\mathbb{P}_{Y|X}^{(k)} = p_\theta(Y|X)$ with the same θ for $k = 0, 1$.

- The MLE on source converges to θ^* regardless of $\mathbb{P}_X^{(1)}$ (as long as it puts mass on the relevant directions).
- Plug in $\hat{\theta}$ to get $\hat{p}_{Y|X}$; this works on the target by assumption.
- Conclusion: **well-specified parametric MLE is automatically robust to covariate shift.**
- *Why?* MLE maximizes $\sum_i \log p_\theta(Y_i | X_i)$ — a conditional loss; the score $\nabla_\theta \log p_\theta(Y | X)$ depends only on $P_{Y|X}$, so the source marginal is irrelevant.
- In fact source-only MLE is *minimax-optimal* on the target under a well-specified model — with *no* bounded-density-ratio assumption (Ge et al., 2024).

When ERM fails: mis-specification or non-parametric

Two regimes where naive ERM is biased on the target:

- **Mis-specified model:** the best linear (etc.) approximation is determined by $\mathbb{P}_X^{(k)}$.
Different $\mathbb{P}_X^{(k)} \Rightarrow$ different “best” parameter $\Rightarrow$ different prediction.
- **Non-parametric estimators** (kernel, nearest neighbor): convergence rates depend on the local density of training points. Source-skewed densities give source-skewed accuracy.

Takeaway: “free” covariate-shift adaptation is the exception, not the rule. When in doubt, correct.

Beyond parametric models: k -NN under covariate shift

Suppose f^* is β -Hölder and $\mathbb{P}_X^{(1)}$ has support strictly containing $\mathbb{P}_X^{(0)}$. Then k -NN regression has target risk

$$R^{(0)}(\hat{h}_{kNN}) - R^{(0)}(f^*) \lesssim \mathbb{E}_{X \sim \mathbb{P}^{(0)}} [w(X)^\beta] \cdot n_1^{-2\beta/(2\beta+d)}$$

where $w(X) = \frac{d\mathbb{P}_X^{(0)}}{d\mathbb{P}_X^{(1)}}(X)$ is the density ratio (recall $w(X) = d\mathbb{P}_X^{(0)}/d\mathbb{P}_X^{(1)}$, the same object reweighting uses) — large exactly where the source is sparse, so the constant $\mathbb{E}[w^\beta]$ (not the rate) is what covariate shift inflates.

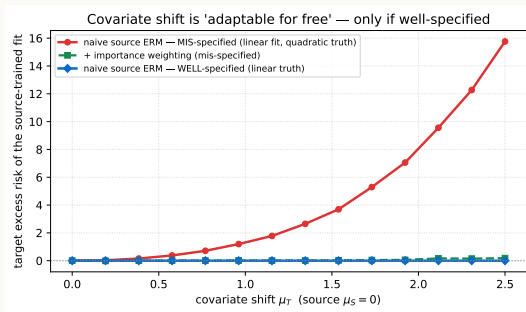
Why w^β ? A β -Hölder f^ makes a neighbor at distance r contribute bias $\sim r^\beta$; where source is sparse (w large) the k -NN radius is large, so bias $\sim w^\beta$ — hence $\mathbb{E}[w^\beta]$.*

- If $\mathbb{P}_X^{(0)}$ concentrates where source is sparse, $\mathbb{E}[w^\beta]$ blows up.
- This is the non-parametric face of covariate shift: the rate is right, the constant explodes.

Fix: reweight source points so they mimic the target density — the reweighting method, next.

“Adaptable for free” — or not: one picture

Sweep the target shift μ_T (source $\mu_S=0$); deploy the source fit *cold* and measure excess risk over the target-optimal fit.



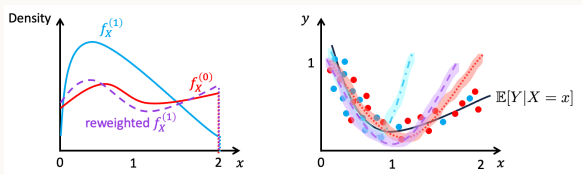
- **Well-specified** (linear truth, linear fit): flat at ≈ 0 for every shift — genuinely free.
- **Mis-specified** (quadratic truth, linear fit): excess risk climbs with the shift — not free.
- True-ratio importance weighting pulls the mis-specified curve back to the oracle.

§3.3: The reweighting method

Reweighted ERM

Fix a loss $\ell(h(X), Y)$ and recall $\mathbb{E}_{\mathbb{P}^{(0)}}[\ell] = \mathbb{E}_{\mathbb{P}^{(1)}}[w(X)\ell]$. Define the **importance-weighted ERM**:

$$\hat{h}_{\text{IW}} = \arg \min_{h \in \mathcal{H}} \frac{1}{n_1} \sum_{i=1}^{n_1} w(\mathbf{x}_i^{(1)}) \ell(h(\mathbf{x}_i^{(1)}), y_i^{(1)}).$$



Source points reweighted by $w(\mathbf{x}_i)$ (larger = “more like target”); after reweighting the empirical law mimics the target.

Software: our group’s R/Python transfer-learning packages implement these methods — https://yangfengstat.github.io/projects/transfer_learning/.

Why reweighting works

Theorem 3.3.1 (Cortes et al., 2010)

The IW objective is unbiased for the target risk, and the deviation is governed by the second-moment / Rényi divergence $d_2(\mathbb{P}^{(0)} \parallel \mathbb{P}^{(1)}) = \mathbb{E}_{\mathbb{P}^{(1)}}[w(X)^2]$: w.p. $1 - \delta$,

$$R^{(0)}(\hat{h}_{\text{IW}}) \leq \min_h R^{(0)}(h) + \mathcal{O}\left(\sqrt{\frac{d_2(\mathbb{P}^{(0)} \parallel \mathbb{P}^{(1)}) \text{VC}(\mathcal{H})}{n_1}}\right).$$

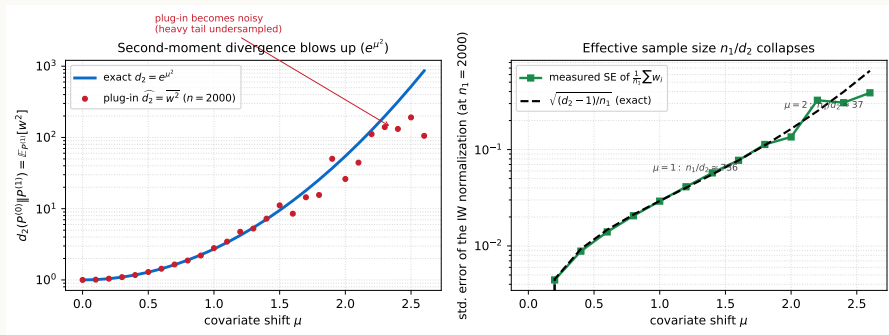
- Unbiased: $\mathbb{E}_{\mathbb{P}^{(1)}}[w(X)\ell(h(X), Y)] = R^{(0)}(h)$.
- *Why?* Under covariate shift $\mathbb{P}_{Y|X}^{(0)} = \mathbb{P}_{Y|X}^{(1)}$, so
$$\mathbb{E}_{\mathbb{P}^{(1)}}[w(X)\ell] = \int w d\mathbb{P}_X^{(1)} \int \ell d\mathbb{P}_{Y|X}^{(1)} = \int d\mathbb{P}_X^{(0)} \int \ell d\mathbb{P}_{Y|X}^{(0)} = R^{(0)}(h).$$

Why reweighting works (cont.)

- “Effective sample size” = n_1/d_2 : heavy-tailed w (large d_2) wastes data. (Bounded $w \leq W$ is the special case $d_2 \leq W$.)
- *Why d_2 ?* The IW risk has variance $\propto \mathbb{E}_{\mathbb{P}^{(1)}}[w^2 \ell^2]/n_1 \leq \mathbb{E}_{\mathbb{P}^{(1)}}[w^2]/n_1 = d_2/n_1$ — noise scales with the *second moment* of the weight, not its max.
- Unbounded w : the rate degrades below $\sqrt{n_1}$; in practice w is unknown $\Rightarrow$ density-ratio estimation (§3.4).

Effective sample size n_1/d_2 , illustrated

Source $N(0, 1)$, target $N(\mu, 1)$: weight $w(x) = e^{\mu x - \mu^2/2}$, and $d_2 = \mathbb{E}_{\mathbb{P}(1)}[w^2] = e^{\mu^2}$ in closed form.



- **Left:** $d_2 = e^{\mu^2}$ blows up super-exponentially; the plug-in $\overline{w^2}$ under-estimates once the heavy tail is undersampled.
- **Right:** the measured s.e. of the IW normalization tracks $\sqrt{(d_2 - 1)/n_1}$ exactly — at $\mu=2$ only $15/33$ $n_1/d_2 \approx 37$ of 2000 points survive.

Check-in: importance weighting

The true density ratio is $w(\mathbf{x}) = e^{2\mathbf{x}_1}$, so $\sup w = \infty$ on $\mathbb{R}^d$. Your source has $n_1 = 10,000$ points.

Which fix is principled?

- (a) Use IW-ERM; the unbiased estimator ensures consistency.
- (b) Clip w at some W , accept a small bias, recover variance control.
- (c) Switch to target-only ERM; we have no labels, so we are stuck.
- (d) Use both methods and take the minimum-MSE estimator.

Answer

(b). Unbounded w blows up the IW variance: a single source point with very large weight can dominate the empirical risk. Truncating at some level W yields a tractable bias-variance trade-off (also justified theoretically when w has heavy tails).

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Numerical illustration: importance weighting

Numerical illustration: setup

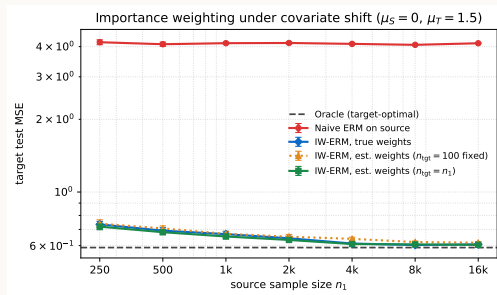
Source $X \sim N(0, 1)$, target $X \sim N(1.5, 1)$ (so $w(x) = e^{1.5x - 1.125}$ [ratio of $N(1.5, 1)$ to $N(0, 1)$ densities, so $w(x) = \exp(1.5x - 1.5^2/2)$], bounded on bounded sets). Truth $y = 0.5x^2 + 0.3\varepsilon$, fit class is *linear* (mis-specified).

- Source OLS: $b \approx 0$ (since $\text{Cov}(x, x^2) = 0$ at $\mu_S=0$).
- Target OLS: $b \approx 1.5$ (best linear fit around $\mu_T=1.5$).

Compare on the target test set against the **oracle** (population target-optimal fit): naive ERM; IW-ERM with *true* weights; and IW-ERM with *estimated* weights (logistic discriminator), using either a small *fixed* unlabeled-target pool ($n_{\text{tgt}}=100$, balanced discriminator) or one that *grows* with n_1 .

Code: `sims/L3_importance_weighting.py`.

Numerical illustration: result



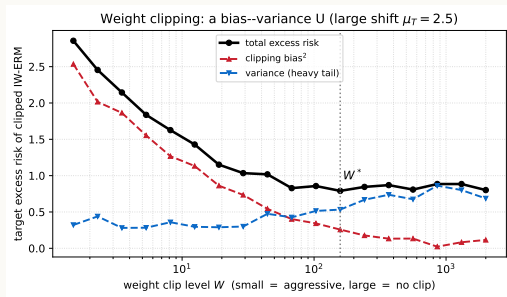
- Naive ERM is biased by $\sim 4\times$ the irreducible target MSE — more source never helps.
- IW-ERM with the *true* weights recovers the oracle as n_1 grows.
- Estimated weights from a *fixed* target pool ($n_{tgt}=100$) plateau *above* the oracle — the weights are only as good as that target sample.
- Grow the target with n_1 and that curve drops to the oracle, matching true weights.

Numerical takeaway

- Under covariate shift + mis-specification, naive ERM is *permanently biased*: more source data does not help.
- IW reweighting is the right correction; it converges to oracle at the standard $\sqrt{n_1}$ rate, scaled by the Rényi divergence $d_2(\mathbb{P}^{(0)} \parallel \mathbb{P}^{(1)}) = \mathbb{E}_{\mathbb{P}^{(1)}}[w^2]$.
- Estimated weights from a binary discriminator work well *given enough unlabeled target* to learn them (previewed in §3.4, the discriminative method).

Weight clipping is a bias–variance dial

Large shift ($\mu_T=2.5$) makes w heavy-tailed. Clip weights at level W , sweep W , and measure the target excess risk of clipped IW-ERM.



- Aggressive clip (small W): weights truncated $\Rightarrow$ **bias** (left wall).
- No clip (large W): the heavy tail survives $\Rightarrow$ **variance** (right wall).
- A U in between: CV-ing W on the target objective lands near W^* — exactly check-in answer (b).

§3.4: Density-ratio estimation

Estimating the density ratio: four canonical routes

Everything so far assumed w known; now we estimate it from source plus unlabeled target features. The four routes are surrogates for the same w : match moments (KMM), classify domains (discriminative), or fit w directly (KL / least-squares).

Given source $\{\mathbf{x}_i^{(1)}\}_{i=1}^{n_1}$ and unlabeled target $\{\mathbf{x}_j^{(0)}\}_{j=1}^{n_0}$:

- **Kernel mean matching** (Huang et al., 2006): match RKHS mean embeddings.
- **Discriminative learning** (Bickel et al., 2009): train a classifier of source vs. target; convert probabilities to a ratio.
- **KL-divergence minimization** (Sugiyama et al., 2008): fit a parametric $\hat{w}$ to maximize the KL surrogate.
- **Least-squares** (Kanamori et al., 2009; Nguyen et al., 2010): solve a convex regression for w directly.

(here n_0 counts unlabeled target points — covariate-shift transfer has no target labels; contrast L2/L4 where n_0 was labeled target examples.)

Density-ratio estimation: scope of the four routes (cont.)

The four routes are the canonical, dimension-robust estimators; the rest of this section develops each in turn.

- **Kernel mean matching** — match RKHS mean embeddings (moments).
- **Discriminative** — classify source vs. target, convert to a ratio.
- **KL / least-squares** — fit w directly via a divergence surrogate.

Other approaches (separate KDE, histograms, semi-parametric) are useful in 1-D / low-dim or under structural assumptions; we summarize them in one slide at the end.

Kernel mean matching (KMM)

Idea: choose source weights $\hat{w}_i$ so that the *reweighted* source mean embedding matches the target mean embedding in an RKHS $\mathcal{H}_K$:

$$\min_{\hat{w}_{\geq 0}} \left\| \frac{1}{n_1} \sum_{i=1}^{n_1} \hat{w}_i K(\mathbf{x}_i^{(1)}, \cdot) - \frac{1}{n_0} \sum_{j=1}^{n_0} K(\mathbf{x}_j^{(0)}, \cdot) \right\|_{\mathcal{H}_K}^2 \quad \text{s.t.} \quad \frac{1}{n_1} \sum_i \hat{w}_i = 1, \quad \hat{w}_i \leq B.$$

- Convex QP in $\hat{w}$; no parametric model for w .
- Connects directly to MMD (Lecture 2): the objective is an empirical squared MMD.
- *Picture*: treat $K(\mathbf{x}, \cdot)$ as a “fingerprint” of $\mathbf{x}$; KMM finds weights so the weighted-average source fingerprint equals the target’s — they look identical to any $f \in \mathcal{H}_K$.

KMM: theory

KMM is an algorithm (Huang et al., 2006); its *consistency* is analyzed later:

Theorem 3.5.1 (Gretton et al., 2009; Yu and Szepesvári, 2012)

With a universal kernel (i.e. $\mathcal{H}_K$ dense in $C(\mathcal{X})$), so matching means forces the full distributions to agree) and bounded weights $\hat{w} \leq B$, the reweighted feature mean converges ($\hat{w} \rightarrow w$ in MMD as $n_1, n_0 \rightarrow \infty$) and the IW-ERM with $\hat{w}$ is consistent.

- Practical note: tune B on a held-out validation set; popular choices $B \in [10, 100]$.
- Strength: works well in moderate dimension; nonparametric.
- Weakness: doesn't extrapolate beyond the source support; sensitive to B .

Discriminative density-ratio learning

Stack source (label $z = 0$) and target (label $z = 1$) and train any binary classifier

$\hat{\eta}(\mathbf{x}) = \widehat{\Pr}(z = 1 \mid \mathbf{x})$. By Bayes, $\Pr(z=k \mid \mathbf{x}) \propto p_X^{(k)}(\mathbf{x}) \Pr(z=k)$, so

$\frac{p_X^{(0)}}{p_X^{(1)}} = \frac{\Pr(z=0 \mid \mathbf{x})}{\Pr(z=1 \mid \mathbf{x})} \cdot \frac{\Pr(z=1)}{\Pr(z=0)}$ (source $z=0$, target $z=1$). Then by Bayes' rule,

$$\frac{\mathbb{P}_X^{(0)}(\mathbf{x})}{\mathbb{P}_X^{(1)}(\mathbf{x})} = \frac{\Pr(z = 1 \mid \mathbf{x})}{\Pr(z = 0 \mid \mathbf{x})} \cdot \frac{\Pr(z = 0)}{\Pr(z = 1)} \approx \frac{\hat{\eta}(\mathbf{x})}{1 - \hat{\eta}(\mathbf{x})} \cdot \frac{n_1}{n_0}.$$

- Any well-calibrated classifier (logistic, GBM, deep net) gives an estimate.
- **Easy to deploy**; the workhorse method in modern practice.
- Extends to non-tabular data (images, text) because the discriminator can.

This is L2's source-vs-target domain classifier, now repurposed to estimate the weight w rather than just measure distance.

Discriminative density-ratio: theory

Theorem 3.5.2 (Bickel et al., 2009, informal)

Suppose the discriminator class is rich enough and trained with $n_1 + n_0$ joint samples; then with high probability,

$$\left(\mathbb{E}_{\mathbb{P}^{(1)}} [(\hat{w} - w)^2]\right)^{1/2} \leq C \sqrt{\text{VC}(\mathcal{F})/(n_1 + n_0)} + \text{approximation error of } \mathcal{F}.$$

- Same rate as plain classification on $z \in \{0, 1\}$ – no curse of dimensionality if the discriminator class is well chosen.
- Couples the IW-ERM analysis to standard classification theory.

KL-divergence minimization (KLIEP)

Parametrize $w_\theta(\mathbf{x}) \geq 0$ (e.g. $w_\theta(\mathbf{x}) = \exp(\theta^\top \phi(\mathbf{x}))$). Maximize

$$\frac{1}{n_0} \sum_j \log w_\theta(\mathbf{x}_j^{(0)}) \quad \text{s.t.} \quad \frac{1}{n_1} \sum_i w_\theta(\mathbf{x}_i^{(1)}) = 1 \quad (\text{enforces } \mathbb{E}_{\mathbb{P}^{(1)}}[w] = 1, \text{ i.e. } \hat{p}_0 \text{ is a density}).$$

Equivalent to minimizing $\text{KL}(\mathbb{P}_X^{(0)} \parallel \hat{w} \cdot \mathbb{P}_X^{(1)})$ over the parametric class.

Sketch: $\text{KL}(p^{(0)} \parallel \hat{w} p^{(1)}) = \mathbb{E}_{p^{(0)}}[\log p^{(0)}] - \mathbb{E}_{p^{(0)}}[\log \hat{w}] - \mathbb{E}_{p^{(0)}}[\log p^{(1)}]$; only the middle term depends on $\hat{w}$, so minimizing KL = maximizing $\frac{1}{n_0} \sum_j \log w_\theta(\mathbf{x}_j^{(0)})$ s.t. normalization.

- **KLIEP** (Sugiyama et al., 2008): closed-form for log-linear w_θ , with cross-entropy-type objective.
- Theory: under standard regularity, $\hat{w}_\theta$ converges at the parametric rate.
- Caveat: parametric form for w may be restrictive, but works well for moderate d .

Least-squares ratio estimation

Direct convex regression for w itself:

$$\min_{w \in \mathcal{F}} \frac{1}{2} \mathbb{E}_{\mathbb{P}^{(1)}} [w(X)^2] - \mathbb{E}_{\mathbb{P}^{(0)}} [w(X)] = \frac{1}{2} \mathbb{E}_{\mathbb{P}^{(1)}} [(w - w^*)^2] + \text{const.}$$

Derivation: $\frac{1}{2} \mathbb{E}_{\mathbb{P}^{(1)}} [w^2] - \mathbb{E}_{\mathbb{P}^{(0)}} [w] = \frac{1}{2} \mathbb{E}_{\mathbb{P}^{(1)}} [w^2] - \mathbb{E}_{\mathbb{P}^{(1)}} [w w^*] = \frac{1}{2} \mathbb{E}_{\mathbb{P}^{(1)}} [(w - w^*)^2] - \text{const}$, using $\mathbb{E}_{\mathbb{P}^{(0)}} [g] = \mathbb{E}_{\mathbb{P}^{(1)}} [w^* g]$.

Estimable by sample averages: $\frac{1}{2n_1} \sum_i w(\mathbf{x}_i^{(1)})^2 - \frac{1}{n_0} \sum_j w(\mathbf{x}_j^{(0)})$.

Theorem 3.5.3 (Kanamori et al., 2009; Nguyen et al., 2010)

For $\mathcal{F}$ a kernel class, $\hat{w}$ converges to w^ in $L^2(\mathbb{P}^{(1)})$ at the kernel-regression rate.*

- Convex; supports kernels and neural-net parameterizations alike.
- Often more robust than KL methods when target has heavier tails than source.

Other approaches (one-slide summary)

Method	Idea	When to use
Separate KDE	Estimate $\hat{p}_0, \hat{p}_1$, take ratio.	Low- d , smooth densities.
Histogram-based	Bin the space; ratio = bin frequency.	Univariate; quick & dirty.
Semi-parametric (Qin '98)	Tilt: $w_\theta = \exp(\theta^\top \phi)$ via empirical likelihood.	Moderate d + structural prior.

Practical guidance.

- Tabular, moderate d : KMM or KLIEP.
- Image / text / large d : discriminative.
- Heavy-tailed target: least-squares.
- Low- d + smooth densities: separate KDE is fine and exploits structure.

Check-in: which method should you pick?

You have 10,000 source images and 1,000 unlabeled target images at $224 \times 224 \times 3$ resolution. You need $\hat{w}$ for IW-ERM on a downstream image classifier.

Which method is the obvious first choice?

- (a) Separate KDE on $224^2 \times 3$ -dimensional pixel space.
- (b) KMM with a Gaussian kernel.
- (c) Train a CNN to classify source vs. target; convert logits to a ratio.
- (d) Histogram on principal components.

Answer

(c), discriminative density-ratio estimation with a deep net. Curse-of-dimensionality kills (a) and (d); KMM is feasible but tuning B and the Gaussian bandwidth in 150,000-D is impractical. A binary classifier directly inherits the deep-net's representation.

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Lecture summary

- Covariate shift = different $\mathbb{P}_X^{()}$, same $\mathbb{P}_{Y|X}^{()}$.
- “Free” adaptation only for well-specified parametric MLE; otherwise ERM is biased.
- Reweight by $w = d\mathbb{P}_X^{(0)} / d\mathbb{P}_X^{(1)}$ to recover the target objective.
- Bounded w + uniform deviation $\Rightarrow$ standard rates with effective sample size n_1/d_2 .
- Density-ratio estimation: KMM, discriminative, KL, least-squares (and friends).

Next: Lecture 4 turns to *posterior drift* – $\mathbb{P}_X^{()}$ is shared but $\mathbb{P}_{Y|X}^{()}$ differs – and asks how to bias an estimator toward the source as a route to “safe transfer.”

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